

Xiaochen Cui

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SUMMARY

Focused on database kernel and system programming. Great at teamwork and English.

EXPERIENCE

- **Shanghai Magus Technology Co., Ltd.** - Chongqing, China

Senior Software Engineer - Database Kernel (Apr 2021 - Apr 2024)

- **Vectorized Query Engine:** Designed and developed the vectorized query engine for LightningDB, which makes SQL queries 50-300 times faster than the original system. (LightningDB is an analytical database.)
Technologies used: Golang, SIMD
- **SQL Optimizer and Planner:** Designed and developed SQL optimizer and planner for LightningDB, and developed a DSL (domain-specific language) for SQL optimization. Reduced the complexity of the SQL optimization module significantly, reduced the complexity of the query engine, and solved 80% of the bugs in the legacy system.
- Designed the structure of data files and index files for the underlying storage, resulting in a 30% reduction in disk footprint and a 100% increase in system IO speed.

- **Burnish Technology** - Chongqing, China

Senior Software Engineer (2019 - 2020)

Software Engineer (2018)

- **High-Performance Gateway:** Designed and developed the IoT gateway, which loads up to 1 million packages per second.
Technologies used: Golang, Redis, Kafka, Cloud-Native Stack
- **Cloud-Native:** Built a complete cloud-native environment, reducing the time to deploy a cluster from three weeks to two days, and the time required to release a new version from one week to one day.
Technologies used: docker, kubernetes, prometheus, grafana, helm, consul
- Built and maintained an Alluxio cluster for the data-analyze team, making them operate data more easily, and also doubles the speed of data processing.
- Done a lot of work for the automation of operations and maintenance.

- **Zhicheng Technology** - Chengdu, China

Junior Software Engineer (2017)

- Backend development, server management.
Technologies used: Python, Django, Twisted, RabbitMQ, Nginx, Linux

OPEN SOURCE PROJECT

- **SmallDB (owner):** A relational database that supports various latch modes, transaction models, and isolation levels.
<https://github.com/small-db/small-db>
- **CockroachDB (contributor):** A open source, cloud-native distributed SQL database.
<https://github.com/cockroachdb/cockroach>

EDUCATION

- **Fairleigh Dickinson University - Vancouver Campus** - Vancouver, Canada

Master of Science in Applied Computer Science (2024 - 2025)

- **Tiangong University** - Tianjin, China

Bachelor of Software Engineering (2013 - 2017)